INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

END-SEMESTER EXAMINATION 2022-23

Academic Session: 2022-2023

Programme: M.Tech. (Computer Science & Engineering) **Semester: Semester I**

Course Code: MCS-101 **Course Title: Advanced Computer Architectu**

Time Allowed: 3 Hours Maximum Marks: 70

Note: Attempt all sections as per instructions. Assume missing data suitably, if any.

SECTION A (Short Answers - 3.5 Marks each)

Q1(a). State Amdahl's Law and calculate speedup with 80% parallelized code on 8 processors.

Q1(b). Explain Flynn's taxonomy classification of computer architectures.

Q1(c). Differentiate between static and dynamic pipeline processors.

Q1(d). Explain the cache coherence problem in multiprocessor systems.

SECTION B (7 Marks each)

Q2. Describe Tomasulo's algorithm for dynamic instruction scheduling with reservation stations.

Q3. Explain branch prediction strategies: 1-bit, 2-bit saturating counters, and correlating branch predictors.

SECTION C (14 Marks each)

Q4. Discuss snooping vs directory-based cache coherence protocols (MESI / MOESI) in detail.

Q5. Explain Vector Processing and SIMD architecture with Matrix Multiplication application.

========== END OF QUESTION PAPER ==========